Lists of Experiments

1. Develop a Python program to calculate the factorial of a number and determine whether a given number is prime using conditional and looping constructs.

2. Develop a Python program to generate the Fibonacci series using recursive and user-defined functions.

3. Develop a Python program to manage student records using lists and dictionaries, supporting insertion, deletion, and search operations.

4. Develop a Python program to analyse student course enrolments by using set operations (union, intersection, difference) to identify students enrolled in common and unique courses, and tuples to store student details.

5. Develop a Python program to count the frequency of words and characters in a text string.

6. Develop a Python program to read student marks from a text file, calculate the total, average, and grade for each student, and write the results to an output file along with their details.

7. Develop a Library Management System using inheritance and polymorphism to manage different types of library items (Book, Magazine, and Journal) and perform issue and return operations.

8. Develop a Python program using NumPy arrays to compute the total marks, average marks, and percentage improvement of students across multiple subjects using vectorized arithmetic operations.

Mid-Semester Examination Marks

Student	Mathematics	Physics	Chemistry	English
S1	65	70	68	72
S2	78	75	80	77
S3	55	60	58	62
S4	82	85	80	88
S5	70	68	72	74

End-Semester Examination Marks

Student	Mathematics	Physics	Chemistry	English
S1	72	78	75	80
S2	84	82	86	83
S3	65	68	66	70
S4	88	90	85	92

S5	78	75	80	82

9. Develop a Python program to compute statistical measures (mean, median, standard deviation, minimum, and maximum) using NumPy and without using NumPy.

marks = [78, 85, 92, 67, 88, 73, 95, 81, 76, 89]

temperatures = [28.5, 30.2, 29.8, 31.4, 27.9, 32.1, 30.5]

sales = [12500, 13800, 14200, 11900, 15100, 14750, 16000]

10. Develop a Python program to perform matrix addition, subtraction, multiplication, and transpose operations using NumPy and without using NumPy.

11. Develop a Python program to compute the determinant, inverse, and rank of a matrix.

12. Develop a Python program to compute eigenvalues and eigenvectors of a matrix.

13. Develop a Python program to perform numerical differentiation and integration of mathematical functions using SciPy.

14. Develop a Python program using SciPy to fit a second-degree polynomial to the following data points and estimate the polynomial coefficients. Plot the original data points and the fitted curve.

x	**y**
1	6
2	17
3	34
4	57
5	86
6	121

(These points approximately follow .)

15. Develop a Python program using SciPy optimization techniques to determine the minimum value of the function.

16. Develop a SQLite-based Student Information Management System using Python to store, retrieve, and display student details such as roll number, name, department, and CGPA.

17. Develop a Python application to perform Create, Read, Update, and Delete (CRUD) operations on employee records containing employee ID, name, designation, department, and salary in an SQLite database.

18. Develop a Python application to generate department-wise employee salary reports by executing parameterized SQL queries on an SQLite database.

19. Develop a Python application with a graphical data-entry form to manage library records, including book addition, issue, return, and search operations using SQLite.

20. Develop a Python program to extract news headlines and hyperlinks from a web page using Beautiful Soup.

24. Develop a Python program to scrape product information from an e-commerce website and store it in CSV format.

22. Develop a Python program to parse and display data from an XML document.

23. Develop a Python program to modify and generate XML documents for student information management.

24. Develop a Python program to clean and preprocess a real-world dataset using Pandas.

25. Develop a Python program using Matplotlib to visualize the monthly sales revenue of a retail store using line charts and bar charts and identify the month with the highest and lowest sales.

Month	**Sales (₹)**
Jan	45000
Feb	52000
Mar	48000
Apr	61000
May	58000
Jun	65000
Jul	62000
Aug	70000
Sep	68000
Oct	75000
Nov	72000
Dec	80000

26. Develop a Python program using Matplotlib to visualize the distribution and relationship between students' study hours and examination scores using histograms and scatter plots.

Student	Study Hours/ Week	Exam Score
S1	8	65
S2	10	72
S3	12	75
S4	15	82
S5	18	88
S6	20	90
S7	7	60
S8	14	80
S9	16	85
S10	22	95

27. Develop a Python program using Seaborn to perform exploratory data analysis on student examination data by visualizing subject-wise score distributions, pairwise relationships, and correlation heatmaps.

Student	Mathematics	Physics	Chemistry	English
S1	78	72	75	80
S2	85	88	82	84
S3	67	70	65	72
S4	90	92	88	91
S5	73	75	78	76
S6	81	79	83	80
S7	69	68	72	70
S8	94	90	92	95
S9	76	74	79	77

S10	88	85	86	89

28. Develop a Python program using Plotly to create an interactive bar chart showing the marks obtained by students in Mathematics.

29. Create a Git repository and perform version control operations including commit, branch creation, and merging.

30. Clone an existing Python project repository, modify a source file, track the modifications using Git commits, create a development branch, and merge the branch into the main branch.

31. Fork an existing GitHub repository, resolve a reported issue, submit the solution through a pull request, and manage repository synchronization using GitHub.